Exercice 07:

$$(S_1): \begin{cases} -mx + z = 1 \\ 2x + my + z = 1 \\ x + y + z = 1 \end{cases}$$

1ère cas: $m = \pm\sqrt{2}$.

$(S_1) \Leftrightarrow A_m X = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ où $X = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$

$$\Leftrightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = A_m^{-1} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \frac{1}{2-m^2} \begin{pmatrix} m-1 & 1 & -m \\ -1 & -m-1 & m+2 \\ -m+2 & m & -m^2 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$\Leftrightarrow \begin{cases} x = \frac{1}{2-m^2}(m-1+1-m) = 0 \\ y = \frac{1}{2-m^2}(-1-m-1+m+2) = 0 \\ z = \frac{1}{2-m^2}(-m+2+m-m^2) = 1 \end{cases}$$

D'où $S_1 = \{(0,0,1)\}$.

2ème cas: $m = \sqrt{2}$

$$(S_1) \Leftrightarrow \begin{cases} -\sqrt{2}x + z = 1 \\ 2x + \sqrt{2}y + z = 1 \\ x + y + z = 1 \end{cases}$$

$$\Leftrightarrow \begin{cases} z = 1 + \sqrt{2}x \\ 2x + \sqrt{2}y + 1 + \sqrt{2}x = 1 \\ x + y + 1 + \sqrt{2}x = 1 \end{cases}$$

$$\Leftrightarrow \begin{cases} z = 1 + \sqrt{2}x \\ y = -(1+\sqrt{2})x \end{cases}$$

D'où $S_2 = \{(x, -(1+\sqrt{2})x, 1+\sqrt{2}x) : x \in \mathbb{R}\}$

$= \{(0,0,1)\} + \mathrm{Vect}((1, -(1+\sqrt{2}), \sqrt{2}))$.

3ème cas : $m = -\sqrt{2}$

$$(S_1) \iff \begin{cases} \sqrt{2}\,x + z = 1 \\ 2x - \sqrt{2}\,y + z = 1 \\ x + y + z = 1 \end{cases}$$

$$\iff \begin{cases} z = 1 - \sqrt{2}\,x \\ y = -x - z + 1 = (-1+\sqrt{2})\,x \end{cases}$$

D'où $S_3 = \{(x, (-1+\sqrt{2})x, 1-\sqrt{2}x) : x \in \mathbb{R}\}$

$= \{(0,0,1)\} + \mathrm{Vect}((1, -1+\sqrt{2}, -\sqrt{2}))$.

$$(S_2) : \begin{cases} 2x + 3y + z = 4 \\ -x + my + 2z = 5 \\ 7x + 3y + (m-5)z = 7 \end{cases}$$

Posons $B_m = \begin{pmatrix} 2 & 3 & 1 \\ -1 & m & 2 \\ 7 & 3 & m-5 \end{pmatrix}$

Calculons $\det B_m = \begin{vmatrix} 2 & 3 & 1 \\ -1 & m & 2 \\ 7 & 3 & m-5 \end{vmatrix}$

$$= \begin{array}{c} \begin{matrix} C_1 - 2C_3 & C_2 - 3C_3 & C_3 \end{matrix} \\ \begin{vmatrix} 0 & 0 & 1 \\ -5 & m-6 & 2 \\ 17-2m & 18-3m & m-5 \end{vmatrix} \end{array}$$

Considère les cas :

- $m \neq 1,\ m \neq 6$ (1ère cas)
- $m = 1$ (2ème cas)
- $m = 6$ (3ème cas)

$= -5(18-3m) - (m-6)(17-2m)$

$= -90 + 15m - 17m + 2m^2 + 102 - 12m$

$= 2m^2 - 14m + 12 = 2(m-1)(m-6)$